

EMX Intelligent Capacity Planning — Evidence-Based Prototype & Blueprint

Prepared for: EMX (Courier, Express and Parcels arm of 7X Group) **Scope:** Workforce capacity planning only — forecast-to-staffing conversion, permanent/outsourced allocation, hiring decisions, rostering support **Date:** 2 August 2026

Source documents (cited throughout by short name and page number):

Short name	Full title	Pages
Doc A	Enterprise Delivery Management and Network Optimisation Platform Blueprint	50
Doc B	EMX Corporate, Commercial, Operational and Strategic Research Profile	53
Doc C	EMX Enterprise Logistics Platform Blueprint	63
Doc D	Blueprint for an Enterprise Delivery Management and Optimization Platform (file: Delivery_Platform_Optimization_Blueprint.pdf)	19

Evidence labels used throughout, per the brief: **Verified** (stated in the brief or PDFs, page-cited) · **Derived** (calculated from a Verified figure via a stated formula) · **Inferred** (reasonable interpretation, not explicit) · **Assumed** (no source figure exists; configurable placeholder) · **Unknown** (cannot be established).

1. Executive Summary

EMX needs to convert a rolling three-month shipment forecast into store-by-store staffing, rostering, hiring and outsourcing decisions. This report and the accompanying reproducible Python prototype (/mnt/user-data/outputs/emx_capacity/) do that, end to end: forecast → workload → productive courier-hours → shrinkage-adjusted headcount → gap → permanent/outsourced allocation → hiring date → explanation.

The single most important finding from the evidence review is that none of the four source documents contain EMX-specific productivity, workforce, cost, or shift-rule data. All three EMX-focused documents state this explicitly and are treated as authoritative on this point:

*"No authoritative EMX-specific figures were identified for revenue, profit, parcel volume, freight tonnage, customer count, **headcount**, market share, fleet size, warehouse capacity... These remain Unknown, not zero."* — Doc B, p.3

*"No authoritative public employee count, **courier count**, nationalisation ratio, functional organisation chart, warehouse staffing level... was identified."* — Doc B, p.36

*"The evidence does not establish EMX shipment volumes, revenue, fleet size, warehouse capacities, route structure, current cost-to-serve, technology vendors, system architecture, service-level performance, **workforce scale**..."* — Doc C, p.1

"Because no company-specific operating data has been supplied, this blueprint assumes a large, multi-hub parcel carrier..." — Doc A, p.1

This is not a gap in this exercise — it is the evidentiary reality the prototype had to be engineered around. Consistent with the brief's explicit instruction not to invent EMX productivity figures, workforce counts, costs, or baseline performance, **every operational parameter in the engine is either (a) taken verbatim from the brief itself — the 60/40 mix, the 45–60 and 5–10 day lead times, the five KPI targets — or (b) an explicitly labelled ASSUMED placeholder**, held in one auditable file (config.py) so it can be swapped for real EMX data without touching any downstream logic. The engine's *structure* — store-level productivity rather than a single network average, shrinkage decomposition, scenario bands instead of point forecasts, a rule layer for structural-vs-temporary gaps feeding a cost-minimising optimiser — is grounded in specific, cited passages from the PDFs (Section 3 and the Evidence Register, Section 20).

What was actually built and run (not just designed): a 12-store synthetic-but-realistically-shaped network, a 7-step capacity-conversion engine, a reconstructed manual baseline, a mixed-integer permanent/outsourced allocator (PuLP), a 26-week time-based backtest with no lookahead, 10 scenario/sensitivity tests, and a store-level explainability generator. All numbers reported in Sections 13–15 are **actual executed output on synthetic, clearly-labelled data** — not illustrative arithmetic and not EMX fact.

Headline backtest result: against a reconstructed manual baseline, the engine cut total mismatch (over+understaffed) courier-hours by **57.5%** (target: 20%) and improved demand-capacity match accuracy from 80.4% to 90.7% (**+10.4 points**, target: 95% — not yet reached on this synthetic dataset, and reported as such). On a fully-costed basis that prices in the cost of reactively closing understaffed hours, the engine reduced cost per shipment by **AED 0.38** (target: AED 0.50 — directionally achieved, not fully met, on synthetic data; see Section 15 for exactly why and how this would be re-tested on real EMX data).

What this is not: a claim that EMX's current planning looks like the reconstructed baseline, or that the specific cost/headcount numbers describe EMX's actual network. It is a working demonstration, on a schema-faithful synthetic dataset, that the proposed architecture is technically sound, reproducible, and measurably better than a flat, store-undifferentiated planning rule — with every assumption laid out for replacement.

2. Verified Understanding of the EMX Challenge

Restating the problem, cross-checked against the evidence:

- Demand changes weekly; permanent capacity changes slowly (45–60 day lead time) — **Verified-Brief**. This structural mismatch is echoed in the PDFs' general logistics framing: Doc A, p.19 lists distinct forecast horizons for distinct decision types (same-day labour movement: hours to 2 days; shift/subcontractor booking: 1–6 weeks; vehicle/facility planning: 1–6 months), which is exactly the multi-horizon planning structure the brief asks for.
- EMX today has a 3-month rolling forecast per store but no systematic forecast→staffing conversion — **Verified-Brief** (no PDF confirms or denies EMX's current planning tooling; Doc B/C confirm no internal process detail is public).
- The 60/40 permanent/outsourced target and the 45–60 / 5–10 day lead times are treated as **Verified-Brief** inputs: exhaustive keyword search of all four PDFs (60%, 40%, workforce mix, lead time, onboarding) found no corroborating or conflicting EMX-specific figure. They are the brief's own operating parameters, not sourced facts, and are used unmodified in the engine.
- EMX's own published scope is far broader than a single planning problem (domestic/international express, freight, fulfilment, cards, autonomous pilots — Doc B, p.1). The brief's instruction to *stay focused on workforce capacity planning and not expand into a general logistics platform* is followed throughout: routing, warehouse, fleet-maintenance and pricing material from the PDFs is used only where it clarifies a capacity-planning input (e.g., productivity formulas), never as a scope target.

Root causes the engine specifically targets, mapped to the brief's own list and to Doc A's general (non-EMX-specific) diagnosis of the same structural problem:

Brief's stated gap	Doc A's generic parallel finding	Engine response
Productivity differences ignored	<i>"Excess vehicle capacity and overtime ← Separate vehicle, driver and shipment allocation"</i> (p.3)	Store-archetype-specific service+travel time (Section 6/8)
Shrinkage (leave, rest days, absence) not modelled	<i>"Capacity forecasting ← Demand forecast plus productivity and absence distributions → Scenario bands, not single-point values"</i> (p.17)	Explicit shrinkage decomposition (Section 8, Step 3)
Lead times not respected	Not addressed in PDFs (EMX-specific)	Hiring-date backward calculation (Section 9)
Reactive, not forecast-led, planning	<i>"Forecast-led labour, vehicle and facility-capacity planning"</i> listed as a highest-value capability (p.3)	Full engine is forecast-led by construction

3. Evidence and Data Audit

3.1 Input inventory

Input	Available?	Source	Granularity	Quality/limitation	Used how	Proxy/assumption needed?
Forecast granularity/horizon	Partially	Brief states "rolling 3-month"	Not specified (day? hour?)	Store/day/hour granularity requested but not described	Engine built to accept and produce day- and hour-level output (Section 8)	Assumed hourly demand curve (no EMX hourly profile published)
Store/geographic data	No	—	—	No EMX store list, count, or location is public (Doc B p.3, p.35, p.41)	Synthetic 12-store network built to the requested schema	Assumed entirely
Shipment volumes/history	No	—	—	Doc B p.35: "Domestic and international parcel volumes: Unknown"	Synthetic history generator (data_gen.py)	Assumed entirely
Courier productivity	No	—	—	Doc B p.36: no courier count or productivity published	Store-archetype service/travel-time parameters	Assumed values; Verified formula structure (Doc C p.38)
Delivery/service times	No EMX figure; formula Verified	Doc C p.38: <i>"Parcels per labour hour = Completed eligible parcel tasks ÷ paid direct labour hours"</i>	—	Generic KPI, not EMX-populated	Used as the engine's productivity denominator structure	Assumed numeric values

Route lengths/travel times	No	—	—	Not published	Archetype travel-time-per-stop parameter	Assumed
Working hours/shift patterns	No	—	—	Not published	8-hour shift Assumed	Assumed
Attendance/leave/absence data	No	—	—	Not published	Flat leave-rate + rest-day-ratio parameters	Assumed
Permanent/outsourced workforce numbers	No	—	—	Doc B p.35: “Headcount and courier count: Unknown”	Synthetic starting workforce, engineered to be moderately under-provisioned and mismatched vs 60/40 (Section 4)	Assumed entirely
Labour/outsourcing costs	No	—	—	No wage or outsourcing-rate figure in any of the four PDFs	AED/hour placeholders (<code>config.py</code>)	Assumed
Courier utilisation	No EMX figure; formula Verified	Doc A p.34: “Driver utilisation = <i>Productive paid time</i> ÷ <i>available paid time</i> ”	—	Formula used; EMX value Assumed at 85% target	Assumed value, Verified formula	
Service-level / OTD data	No EMX figure; formula Verified	Doc C p.38: “On-time delivery = <i>Delivered within product commitment</i> ÷ <i>eligible delivered shipments</i> ”	—	Formula referenced in KPI framework (Section 15)	No EMX baseline OTD exists to compare against	
Store operating hours	No	—	—	Not published	08:00–22:00 Assumed	Assumed
Recruitment/onboarding lead times	Verified-Brief	Brief: 45–60 days permanent, 5–10 days outsourced	Point figures	Directly usable	Used verbatim in hiring-date logic	None needed
Constraints/policies/business rules	Partial	60/40 mix target (Verified-Brief); general decision-governance framework (Doc A p.2: four decision classes — automated / planner-approved / rules-based / executive)	—	Governance framework informs the human-approval workflow (Section 10)	Applied to override/approval design	—

3.2 Dataset quality assessment

No operational dataset was attached to this brief — only the four PDF blueprints/research profiles. Consequently the standard checks (missing values, duplicates, inconsistent timestamps, outliers, data leakage, insufficient history, aggregation mismatches) were run **on the synthetic dataset generated to exercise the prototype**, as a demonstration of the checks a production ingestion pipeline must run, not as a finding about EMX data:

- **Missing values:** none by construction (synthetic generator fills every store-day).
- **Duplicates:** checked via `(store_id, date)` uniqueness constraint in `data_gen.py`; none found.
- **Timestamp consistency:** single daily grain, UTC-naive dates; a production pipeline must reconcile UAE local time, DST-equivalent absence (UAE has none, but GCC cross-border ops may differ), and store-open-hours boundaries.
- **Outliers:** synthetic noise is Normal(0, 9%) per store-day; a production pipeline needs an explicit outlier-detection step (e.g., IQR or seasonal-hybrid ESD) before either the forecast or the trailing-average baseline consumes raw actuals — this is flagged as a build item in Section 17, not yet implemented in the prototype.
- **Data leakage:** actively guarded against in the backtest (Section 13) — the naive seasonal forecast and the reconstructed baseline use only data strictly before each evaluation date; the “true required headcount” benchmark (built from realised actuals) is used only for ex-post scoring, never fed back into the forecast.
- **Insufficient history:** the backtest reserves a 28-day warm-up before scoring begins, exactly because the naive seasonal forecast needs same-weekday history; a cold-start / new-store scenario is tested separately (Section 14).
- **Aggregation mismatches:** the engine carries shipments → minutes → hours → headcount through explicit unit-labelled columns at every step

(Section 8) specifically to prevent silent unit errors when real EMX data (which may arrive at order-line, shipment, or manifest grain) is substituted in.

4. Key Assumptions, Uncertainties, and Data Gaps

This is the single most consequential section for anyone deciding whether to trust the numbers in Sections 13–15. Every numeric parameter feeding the engine is declared in one file, `config.py`, each tagged with its evidence status. The full table is reproduced here; **replacing this table with real EMX figures is the single highest-priority pre-pilot task** (Section 17, Phase 1).

Parameter	Value used	Status	Why this value
Permanent share target	60%	Verified-Brief	Stated directly
Outsourced share target	40%	Verified-Brief	Stated directly
Mix tolerance band	±5 pts	Assumed	Operational flexibility band; not specified
Permanent lead time	45–60 days (52.5-day midpoint used in point calcs)	Verified-Brief	Stated directly
Outsourced lead time	5–10 days (7.5-day midpoint)	Verified-Brief	Stated directly
Structural-gap confirmation window	4 consecutive weeks	Assumed	No EMX threshold exists; configurable
Shift length	8.0 hours	Assumed	Standard single-shift placeholder
Rest-day ratio	1/7	Assumed	One rest day per week placeholder
Leave/absence rate	8%	Assumed	Placeholder; no EMX HR data
Break ratio	10% of shift	Assumed	Placeholder
Training/non-productive ratio	3% of shift	Assumed	Placeholder
Utilisation target	85%	Assumed	Placeholder; formula source Verified (Doc A p.34)
Permanent cost/hour	AED 22.00	Assumed	No EMX wage data exists anywhere in the PDFs
Outsourced cost/hour	AED 27.00	Assumed	Same
Emergency outsourced premium	×1.35	Assumed	Placeholder for <5-day emergency booking
Hiring cost/permanent head	AED 3,500	Assumed	Placeholder recruitment+onboarding cost
Service time by archetype	2.6–4.2 min/shipment	Assumed value / Verified formula structure	Formula: Doc C p.38
Travel time by archetype	2.0–4.8 min/shipment	Assumed	No EMX route data
Hourly demand shape	Stylised peak-AM/peak-PM curve	Assumed	No EMX hourly profile published
Weekend uplift	×1.18	Assumed	UAE Fri–Sat weekend pattern is Verified as the general convention; the multiplier itself is Assumed
Campaign uplifts (Ramadan/Eid/BF/year-end)	×1.12 to ×1.60	Assumed value / Verified qualitative signal	Doc A p.1 names these peaks qualitatively but does not quantify them
Forecast uncertainty (CV)	12% at 1 week, +1.5pp/week	Assumed	Structure (scenario bands) is Verified (Doc A p.17); numeric spread is not

Named uncertainties and their downstream impact:

1. **Existing workforce baseline is entirely synthetic.** The single biggest driver of every headcount-gap number in this report. Section 13's backtest results (accuracy improvement, mismatch-hour reduction) are **robust to this** because they compare two *methods* (baseline heuristic vs. engine) applied to the *same* synthetic actuals — the relative improvement is informative even though the absolute headcount numbers are not EMX facts. Section 15's absolute cost figures are **not** robust to this and must be re-run on real payroll/roster data before being used in a business case.
2. **60/40 mix definition is ambiguous in the brief** (by headcount? hours? store? network?) — addressed explicitly in Section 9.

3. **No documented current manual planning formula exists** — the baseline in Section 5 is explicitly a *reconstruction*, not a description of EMX’s actual practice, which remains **Unknown**.
4. **Store archetype productivity values are placeholders** built to be internally *consistent* (denser stores → shorter travel time, more stops/hour) but not benchmarked against EMX telemetry, since none is public.

5. Manual Baseline Definition (Reconstructed)

Per the brief’s own instruction — “Where no documented manual formula exists, define a transparent baseline based on the simplest defensible current-state rule and label it as a reconstructed baseline” — the following reconstruction is used, deliberately built to mirror the failure modes the brief itself describes (“These decisions do not consistently account for different productivity levels across stores... shift structures... hiring and onboarding lead times”):

Reconstructed Manual Baseline formula:

```
trailing_avg_shipments(store) = mean(actual_shipments) over the prior 28 days, as of the planning date
baseline_required_headcount(store) = trailing_avg_shipments / (55 parcels/courier/day × 0.85 buffer factor)
```

Deliberate design choices that make this a faithful reconstruction of *manual, spreadsheet-driven* planning as generically described in the industry literature reflected in the PDFs, not a straw-man:

- **One flat network-wide productivity constant (55 parcels/courier/day)** applied to every store regardless of archetype — this is exactly the anti-pattern the brief calls out (“Do not rely on a single average productivity value across all stores”).
- **No hour-level or weekday/weekend view** — a single daily headcount number per store.
- **No shrinkage decomposition** — a single flat 0.85 “buffer,” not leave/rest/breaks/utilisation modelled separately.
- **Reactive hiring** — outsourced capacity is added only after a shortfall is already visible in the trailing average; no lead-time-aware trigger, and permanent headcount is never proactively adjusted by the baseline rule.
- **No forecast use at all** — a trailing average of the past, not the rolling 3-month forecast, consistent with the brief’s description of current practice as estimation rather than forecast-driven.

This baseline is implemented in `baseline.py` and is what the engine is measured against throughout Sections 13–15.

6. Capacity-Planning Methodology

6.1 Planning problem definition

Element	Definition used
Planning horizon	Rolling 90 days (Verified-Brief : “rolling three-month shipment forecast”)
Decision frequency	Daily refresh of the 90-day plan; weekly review cadence for hiring/outsourcing decisions (aligned to Doc A p.19’s “shifts and subcontractor booking: one to six weeks” horizon)
Planning granularity	Store × date × hour (day-level engine run, with an hourly-breakdown expansion — Section 8)
Demand unit	Forecast shipments (parcels) per store per day
Capacity unit	Productive courier-hours; secondarily, courier headcount
Decision variables	New permanent headcount per store; new outsourced headcount per store (structural); peak outsourced booking per store (temporary); hiring/booking dates
Operational constraints	Shift length; rest-day/leave shrinkage; utilisation ceiling/floor; permanent/outsourced lead times; 60/40 network mix (soft)
Cost components	Scheduled labour cost (permanent + outsourced hourly rates); emergency-outsourced premium for uncovered shortfall; hiring/onboarding cost; overtime premium (modelled, not yet exercised in the base scenario)
Service-level requirement	On-time delivery target 95% (Verified-Brief) — used as a downstream KPI, not a direct engine constraint in this prototype (see Section 15 for how it would be wired in with real OTD-vs-staffing data)
Utilisation boundaries	70% floor / 95% ceiling (Assumed)
Hiring/outsourcing lead-time rules	45–60 / 5–10 days (Verified-Brief)
Mix rule	60/40 target ± 5pp, network-wide, by headcount, soft-constrained (Section 9)

Shortage/overtime/transfer/outsourcing/hiring triggers	Rule layer in Section 9
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6.2 The five capacity concepts, explicitly distinguished

Concept	Where computed	Formula
Forecasted demand	<code>forecast.py</code> output / <code>data_gen.build_forecast</code>	Shipments per store-day, with P50/P80/P95 bands
Required productive courier-hours	<code>engine.required_courier_hours</code>	$\text{shipments} \times (\text{service_min} + \text{travel_min}) / 60$
Scheduled courier-hours	<code>engine.apply_shrinkage_and_capacity</code>	Restated equal to required (Step 5) — this is what should be scheduled, before shrinkage is applied to convert to headcount
Available courier-hours	$\text{effective_hours_per_head} \times \text{existing_headcount}$	Shift hours $\times$ availability factor $\times$ productive fraction $\times$ heads
Effective capacity after shrinkage	Same as available courier-hours	—
Required headcount	<code>engine.apply_shrinkage_and_capacity</code>	$\text{required_scheduled_hours} / \text{effective_hours_per_head}$
Existing headcount	Workforce data	permanent + outsourced on roster
Staffing gap	<code>engine.staffing_gap</code>	$\text{required_headcount} - \text{existing_headcount}$
Hiring requirement	<code>optimize.py</code>	Section 9

7. AI and Optimisation Architecture

7.1 Selected approach and justification

A **hybrid system** — deterministic rules + statistical/time-series forecasting + mixed-integer optimisation + an explainability layer — was selected over a single end-to-end ML model, directly following the design principle stated in Doc A, p.16:

“Use deterministic rules for legal eligibility, safety, contractual commitments and simple workflow. Use mathematical optimisation for allocation, routing, loading, network flow and scheduling. Use machine learning where the decision depends on uncertain outcomes such as demand...”

Layer	What it does	Method	Why this method	Data needed	Calibration	Leakage control	Cold-start handling	Explainability
Demand forecasting	Shipments per store/day	Hierarchical time-series (network $\rightarrow$ city $\rightarrow$ store), with day-of-week and campaign regressors — structure per Doc A p.17 (“Hierarchical time-series, gradient boosting, event regressors”)	Store volumes share network-level seasonality (campaigns, weekend pattern) but have store-specific levels; hierarchical models borrow strength across stores, which matters for low-volume/new stores	≥ 1 year history per store for full seasonality; network-level model can serve new stores immediately	Time-based cross-validation, refit weekly	Strict train/test split by date; naive-seasonal forecast in the backtest only ever sees data before the evaluation date	New stores default to the archetype/city hierarchical prior until they accumulate their own history (demonstrated in Section 14)	Forecast decomposition: trend/seasonal/cycle components, shrinkage factors
Capacity conversion	Demand $\rightarrow$ required headcount	Deterministic formula chain (Section 8)	Fully auditable; the brief requires “explicit” calculations, which rules out a black-box model here	Store archetype parameters, shrinkage rates	Re-estimated whenever HR/roster data changes	N/A (deterministic)	Uses archetype default until store-specific productivity is observed	Every term in the model is named, inspected in the column

Structural vs. peak classification	Separates persistent gaps (→ permanent) from temporary spikes (→ outsourced)	Rule layer on weekly-aggregated required headcount (median vs. 90th percentile)	Brief explicitly requires <i>not</i> recommending hiring for every short-term shortage; a transparent statistical rule is more auditable to planners than a learned classifier for this binary, high-stakes decision	90-day forecast panel	Threshold (4 consecutive weeks) is planner-tunable	N/A	New stores inherit archetype-level structural/peak split until their own pattern is observable	Rule and threshold every explanation 10)
Permanent/outsourced allocation	Minimises network cost subject to mix and coverage constraints	Mixed-integer program (PuLP/CBC)	Brief explicitly asks for MIP; allocation is a small, well-structured combinatorial problem well suited to exact solving at this scale (12–100s of stores)	Structural gap per store, cost rates, mix target	Re-solved on every planning cycle	N/A	New store's gap enters the same MIP as any other store	Solver objective, and realised mix alongside every : (optimisation_
Forecast uncertainty → staffing	P50/P80/P95 bands inform buffer sizing	Parametric CV-based bands, widening with horizon	Matches Doc A p.17's explicit instruction that capacity forecasting output should be "scenario bands, not single-point values"	Forecast residual history (for a calibrated CV; currently an Assumed placeholder CV)	Recalibrate CV from realised forecast error each cycle	Same as forecasting layer	Wider bands by default for new stores (higher assumed CV)	Bands shown also point plan (Secti

Why not a single end-to-end learned optimiser: the brief requires calculations to be “explicit,” recommendations to be explainable in a specific per-store narrative format, and hiring/outsourcing decisions to be auditable by planners who are not data scientists. A fully learned system would satisfy none of these without an explanation layer bolted on afterward, and — critically — cannot be checked against the fact that most of its training signal (EMX-specific productivity, cost, workforce data) does not exist yet. The hybrid design lets every stage be validated independently as real data becomes available, rather than requiring a large labelled dataset upfront.

7.2 Forecast uncertainty and its staffing consequence

Rather than plan to a single-point forecast, the engine can be run against P50, P80, or P95 shipment volumes (`forecast.py` output). The “Forecast deteriorates” scenario in Section 14 runs the whole network at P95 and shows the resulting headcount and cost impact explicitly, rather than asserting robustness qualitatively.

8. Mathematical Formulation and Calculation Logic

All nine required calculation steps, exactly as implemented in `engine.py` (function names shown for traceability):

Step 1–2 — Forecast workload → productive courier-hours (`engine.required_courier_hours`)

```
workload_minutes(store, date) = shipments(store, date) * (service_min[archetype] + travel_min[archetype])
required_courier_hours(store, date) = workload_minutes / 60
```

Step 3 — Capacity shrinkage (`engine.apply_shrinkage_and_capacity`)

```
availability_factor      = 1 - rest_day_ratio - leave_rate                                     (headcount-availability side)
productive_hour_fraction = (1 - break_ratio - training_ratio) * utilisation_target          (hour-productivity side)
effective_hours_per_head = shift_hours * availability_factor * productive_hour_fraction
```

With the current (Assumed) placeholders: $1 - 1/7 - 0.08 = 0.777$ availability $\times (1 - 0.10 - 0.03) \times 0.85 = 0.740$ productive fraction $\times 8$ hours = **4.60 effective productive hours per employed courier per day** — i.e. roughly 58% of a nominal 8-hour shift converts to productive delivery time once rest days, leave, breaks, training and a realistic (not 100%) utilisation target are stacked. This compounding is exactly why the brief calls out shrinkage as a first-class planning input rather than an afterthought.

Step 4–5 — Effective available / required scheduled courier-hours

```
effective_available_courier_hours(store) = existing_headcount(store) × effective_hours_per_head
required_scheduled_courier_hours(store, date) = required_courier_hours(store, date) [restated]
```

Step 6 — Required headcount

```
required_headcount(store, date) = required_scheduled_courier_hours / effective_hours_per_head
```

Step 7 — Staffing surplus/shortage (engine.staffing_gap)

```
staffing_gap = required_headcount - existing_headcount
gap_direction = "understaffed" if gap > 0.25, "overstaffed" if gap < -0.25, else "balanced"
```

Step 8 — Permanent/outsourced allocation — mixed-integer program (optimize.optimise_permanent_outsourced_mix), formulated as:

```
minimise  Σ_s [ perm_add[s] · C_perm + outs_add[s] · C_outs ] + λ · (slack_hi + slack_lo)

subject to  perm_add[s] + outs_add[s] ≥ structural_gap[s]                ∀ store s      (coverage)
            (Σ_s existing_perm[s] + perm_add[s]) ≥ (target-tol) · T - slack_lo    (mix, soft lower)
            (Σ_s existing_perm[s] + perm_add[s]) ≤ (target+tol) · T + slack_hi    (mix, soft upper)
            perm_add[s], outs_add[s] ∈ ℤ≥0

where  T = Σ_s (existing_total[s] + structural_gap[s]) [linearisation constant, see note below]
        C_perm = permanent_cost_per_hour × shift_hours × 260 working days (annualised)
        C_outs = outsourced_cost_per_hour × shift_hours × 260 working days
        λ = AED 50,000 (mix-violation penalty; deliberately high so the model only
            breaches the mix band when coverage would otherwise be infeasible or
            when doing so is unambiguously cheaper than the penalty)
```

Linearisation note (Derived, disclosed): the total-headcount term τ in the mix constraint is technically itself a decision-dependent quantity (existing + all additions). To keep the model a linear (not quadratic) program, τ is approximated using the *minimum feasible* additions (existing + structural gap) before solving, and the **realised** mix ratio is recomputed and reported after solving (optimization_meta.json) so any discrepancy from the target band is visible rather than hidden. In the executed run this produced a realised network permanent share of 64.8% against a 60% ± 5 pp target — inside the band, so no violation occurred in this instance (Section 9.3).

Step 9 — Hiring requirement and hiring date (optimize.hiring_and_outsourcing_dates)

```
latest_permanent_hire_start_date = structural_need_date - 52.5 days (midpoint of 45–60)
permanent_hire_available_date    = today + 52.5 days
outsourced_booking_date          = today
outsourced_available_date        = today + 7.5 days (midpoint of 5–10)
```

Because structural gaps are, by construction, already present at the median week of the 90-day forecast (not a future event), the “latest start date” calculation correctly outputs a date at or before today for every store with a structural gap in the executed run — the engine flags these explicitly as **overdue/immediate hiring actions** rather than silently back-dating them (Section 9.2 shows the full table).

8.1 Hourly breakdown

To satisfy the “by store, date, shift, and hour” requirement, engine.hourly_breakdown distributes each store-day’s required courier-hours across operating hours using an **Assumed** stylised intraday demand curve (08:00–22:00, peaking 10–11 and 16–19). A one-week sample for all 12 stores is in engine_hourly_capacity_plan_week1_sample.csv (84 store-days $\times$ 14 hours = 1,176 rows). **This is the weakest-evidenced part of the engine** — no EMX hourly pattern is published anywhere in the four PDFs — and is flagged as the top candidate for early real-data substitution (Section 17).

9. Permanent, Outsourced, Rostering, and Hiring Decision Rules

9.1 Structural vs. temporary classification

For each store, the 90-day forecast is aggregated to weekly average required headcount. The **structural level** is the median weekly requirement; the **peak level** is the 90th percentile. This directly implements the brief’s instruction “Hiring should not be recommended for every short-term shortage... establish rules... that determine when a sustained gap justifies permanent recruitment rather than outsourcing”:

- **Structural gap** ($\text{structural_level} - \text{existing_headcount}$, floored at 0) → candidate for **permanent** hiring, because by definition it persists across at least half the 13-week horizon — well beyond even the long end of the outsourced lead time, and long enough that permanent hiring (45–60 days) lands with capacity to spare.
- **Peak gap** ($\text{peak_level} - \text{structural_level}$, floored at 0) → candidate for **outsourced-only** coverage, because it is by construction a minority-of-weeks phenomenon that does not justify carrying permanent headcount.

Disclosed limitation: in the executed run, the peak gap came out at almost exactly 1 head-equivalent for every store (Section 8’s “hourly breakdown” caveat applies here too, for a related reason) — because the current forward-looking forecast is a smooth point estimate (deterministic trend + weekend/campaign multipliers, no injected day-to-day noise; uncertainty is instead carried separately in the P50/P80/P95 bands), the week-to-week variation captured by the structural/peak split is dominated by ISO-week boundary effects rather than genuine demand volatility. **A production version should compute the peak component from either (a) actual day-level demand volatility once real history exists, or (b) the gap between the P50 and P80/P95 uncertainty bands**, not from a week-averaged point forecast. This is flagged explicitly rather than presented as a validated volatility measure.

9.2 Store-level recommendations (executed prototype output, synthetic data)

Store	Archetype	Existing HC	Structural level	Peak level	Structural gap	Recommended permanent	Recommended outsourced (structural)	Recommended outsourced (peak)	Permanent hire action date
AAN-01	mixed	4	5.34	5.46	2	2	0	1	02 Aug 2026 (overdue → act now)
AUH-01	urban dense	6	10.35	10.57	5	5	0	1	02 Aug 2026
AUH-02	suburban	6	10.50	10.73	5	5	0	1	02 Aug 2026
AUH-03	industrial b2b	7	10.97	11.21	4	0	4	1	— (outsourced only)
DXB-01	urban dense	9	13.37	13.66	5	1	4	1	02 Aug 2026
DXB-02	urban dense	8	11.65	11.90	4	0	4	1	— (outsourced only)
DXB-03	suburban	7	12.30	12.57	6	6	0	1	02 Aug 2026
DXB-04	industrial b2b	8	12.87	13.15	5	5	0	1	02 Aug 2026
FUJ-01	suburban	3	4.50	4.60	2	0	2	1	— (outsourced only)
RAK-01	mixed	3	4.92	5.03	2	2	0	1	02 Aug 2026
SHJ-01	suburban	7	11.70	11.95	5	5	0	1	02 Aug 2026
SHJ-02	mixed	5	8.44	8.62	4	4	0	1	02 Aug 2026
Network total		73			49	35	14	12	

The solver chose outsourcing-only for AUH-03, DXB-02 and FUJ-01 even though each has a structural (persistent) gap — this is the MIP correctly trading off network-wide mix compliance against per-store cost: adding more permanent heads at every structural-gap store would have pushed the network permanent share above the 65% upper band, so the optimiser filled some structural gaps with outsourced capacity instead to keep the realised mix (64.8%) inside the target band, at a small serviceability cost (outsourced capacity for a persistent need carries repeat 5–10 day booking risk that permanent capacity does not — flagged in Section 18).

9.3 The 60/40 mix: definition, interpretation, and when to break it

The brief does not specify whether 60/40 should be evaluated by headcount, courier-hours, shipment capacity, per-store, per-region, or network-wide — this ambiguity is stated explicitly rather than resolved silently, per the brief’s own instruction.

Definition selected for this prototype: network-wide, by headcount, evaluated after recommended actions. Justification: (a) headcount is the only unit both permanent and outsourced capacity share unambiguously without a conversion assumption (courier-hours would require assuming outsourced couriers work the same shift pattern, which is not established); (b) network-wide (not per-store) avoids forcing small stores (e.g., FUJ-01 with 3 existing heads) into an impossible discrete ratio — a single-head store literally cannot be “60% permanent.”

Conditions under which strictly enforcing 60/40 would create avoidable cost or service risk, tested directly via the mix-sensitivity scenario (Section 14):

Permanent target	Total recommended additions cost (AED, annualised)	Realised permanent share	Δ cost vs. 60% target
50%	2,512,640	54.9%	−124,800 (cheaper)
60% (brief target)	2,387,840	64.8%	—
70%	2,263,040	74.6%	−124,800 (cheaper)

Because permanent cost/hour (AED 22) is *lower* than outsourced (AED 27) in the placeholder cost assumptions, pushing the mix further toward permanent is monotonically cheaper on pure scheduled-labour cost in this synthetic run — the 60/40 target is therefore not being driven by scheduled-cost minimisation at all, but by **flexibility and lead-time risk**: a network at 100% permanent would be unable to respond to the “demand +30%” or “sustained campaign” scenarios (Section 14) within the horizon, since permanent hiring takes 45–60 days. The genuine trade-off the 60/40 target is managing is **cost-of-flexibility vs. cost-of-labour**, not a trade-off visible in scheduled labour cost alone — this is stated explicitly because a reader skimming only the cost table above could otherwise wrongly conclude the mix target is costing EMX money.

9.4 Explicit trigger rules

Condition	Recommended action
Structural gap > 0, confirmed for ≥4 consecutive weeks, need date ≥ permanent lead time away	Permanent hire (subject to network mix constraint)
Structural gap > 0, need date < permanent lead time away (i.e., already overdue, as in every store above)	Flag as overdue hiring action ; cover the interim with outsourced/emergency capacity while recruitment proceeds
Peak gap only (temporary)	Outsourced booking, timed to the 5–10 day lead time
Gap driven by a single low-productivity period, not sustained	Do not hire; escalate as a productivity/training or shift-design investigation (rule present in design, not yet triggered in the executed scenarios since synthetic productivity is stationary per store)
Absence-driven shortfall spike	Short-term outsourced/overtime bridge, not permanent hiring
Network mix would breach ±5pp band	Soft-constraint penalty in the MIP; violation reported explicitly if it occurs

10. Prototype Design and Planner Workflow

10.1 Explainability — actual model output, not template text

Per the brief’s required format, explanations are generated from the live model state (`explain.py`), not from a fill-in-the-blank template. Three real examples from the executed run:

DXB-01 (urban dense) is expected to require 71 productive courier-hours on its peak day in the 90-day window (30 October 2026, 930 forecast shipments). After rest days, leave, breaks and target utilisation are applied, its 9 existing couriers can supply about 41 hours, a gap of 30 hours on that day. Across the full 90-day horizon this store’s requirement stays persistently above its current headcount (structural gap of 5 head-equivalents at the median week), so 1 permanent courier(s) and 4 outsourced courier(s) are recommended to close it. Recruitment for the permanent addition(s) should begin no later than 02 August 2026 given the 45–60 day hiring lead time; this date is already at or before today, so the recommendation is flagged as an immediate/overdue hiring action. In addition, demand volatility adds a temporary peak of about 1 extra head-equivalent above the structural level in the busiest weeks; this is recommended to be covered with short-lead-time outsourced capacity (5–10 day booking) rather than permanent hiring, since it does not persist.

DXB-04 (industrial b2b): required 69 hours on its peak day (30 Oct 2026, 458 shipments); 8 existing couriers supply ~37 hours, a 32-hour gap. Structural gap of 5 heads → 5 permanent, 0 outsourced recommended; hiring flagged overdue.

AAN-01 (mixed): required 29 hours on its peak day; 4 existing couriers supply ~18 hours, a 10-hour gap. Structural gap of 2 heads → 2 permanent recommended; hiring flagged overdue.

10.2 Planner interface — functional specification

Screen/panel	Content	Data source
Network capacity outlook	90-day required vs. existing headcount, network-wide	<code>engine_daily_capacity_plan.csv</code> , aggregated
Store shortages/surpluses	Sortable table, <code>gap_direction</code> flag	<code>engine.staffing_gap</code> output

Demand-capacity heatmap	Store × hour grid	hourly_breakdown output
Permanent/outsourced capacity view	Existing vs. recommended, per store	store_hiring_recommendations.csv
Recruitment pipeline & deadlines	Hiring action dates, overdue flags	hiring_and_outsourcing_dates output
Stores at risk	gap_direction == "understaffed" beyond a configurable threshold	Engine output
Cost/utilisation impact	Scheduled + full-coverage cost per scenario	Section 15 KPI outputs
Forecast uncertainty	P50/P80/P95 bands per store	forecast.py output
Recommendation explanations	Per-store natural-language text	explain.py
Scenario controls	Toggle any Section 14 scenario	scenarios.py
Manual override + audit history	Planner can override any recommendation; override, reason, and timestamp logged	Specified in Section 16 (not yet built — planner-facing UI is out of scope for this prototype, which is engine + API-ready output)

What was actually built vs. specified: the calculation engine, optimiser, backtest, scenario runner and explanation generator are executable Python (Section 12). The planner-facing interface above is a **functional specification**, not a built UI — building a production interface was out of scope for “workforce capacity planning” per the brief’s own scope guardrail, and the engine’s output is deliberately structured (flat CSV/JSON with named columns) so it can be wired directly into a lightweight internal tool or an existing planning UI without redesign.

11. Data Model and Required Fields

Entity	Key fields	Grain	Notes
store	store_id, city, archetype, service_min, travel_min, stops_per_hour_cap, existing_permanent, existing_outsourced	1 row/store	Archetype-based productivity is a placeholder for real per-store telemetry
forecast	store_id, date, forecast_p50, forecast_p80, forecast_p95, forecast_cv, is_weekend, campaign	1 row/store/day	Matches Doc A p.17’s “scenario bands, not single-point values”
history_actuals	store_id, date, actual_shipments, campaign, is_weekend	1 row/store/day	Used only for ex-post backtest scoring and baseline trailing-average
engine_daily_capacity_plan	store_id, date, shipments, workload_minutes, required_courier_hours, effective_hours_per_head, required_headcount, existing_headcount, staffing_gap, gap_direction	1 row/store/day	Full audit trail of Steps 1–7
engine_hourly_capacity_plan	store_id, date, hour, required_courier_hours_hour, shipments_hour	1 row/store/day/hour	Hour-level expansion
store_hiring_recommendations	store_id, structural_level, peak_level, structural_gap_int, peak_gap_int, recommended_permanent_add, recommended_outsourced_add, permanent_hire_action_date, outsourced_booking_date	1 row/store	Output of Section 9’s optimisation
backtest_detail	store_id, date, engine_required_headcount, true_required_headcount, baseline_required_headcount, engine_error, baseline_error, ...	1 row/store/backtest-day	Full validation trail (Section 13)

This schema mirrors what a real EMX extract would need to supply: **stores, a rolling forecast with uncertainty, actuals for backtesting, and a roster (existing_permanent/existing_outsourced per store)**. Because the engine only depends on this schema — not on the synthetic generator — swapping

in real EMX exports requires no change to `engine.py`, `optimize.py`, `backtest.py` or `scenarios.py`.

12. Reproducible Prototype Code

Fully executed, not illustrative. All code below (and the complete files) ran successfully end-to-end in this environment; every number in Sections 9, 13, 14 and 15 is this code's actual output on the synthetic dataset described in Section 3–4, not hand-calculated or invented. Full files are provided as downloadable attachments; the core formula implementation is reproduced here for direct review:

```
# engine.py – Steps 1-7 (core formulas)
def workload_minutes(row):
    return row["shipments"] * (row["service_min"] + row["travel_min"])

def required_courier_hours(df):
    df["workload_minutes"] = df.apply(workload_minutes, axis=1)
    df["required_courier_hours"] = df["workload_minutes"] / 60.0
    return df

def apply_shrinkage_and_capacity(df, absence_rate_override=None):
    leave_rate = cfg.LEAVE_RATE if absence_rate_override is None else absence_rate_override
    availability_factor = (1 - cfg.REST_DAY_RATIO - leave_rate)
    productive_fraction = (1 - cfg.BREAK_RATIO - cfg.TRAINING_NONPRODUCTIVE_RATIO) * cfg.UTILISATION_TARGET
    effective_hours_per_head = cfg.SHIFT_HOURS * availability_factor * productive_fraction
    df["effective_hours_per_head"] = effective_hours_per_head
    df["required_scheduled_courier_hours"] = df["required_courier_hours"]
    df["required_headcount"] = df["required_scheduled_courier_hours"] / effective_hours_per_head
    return df

def staffing_gap(df):
    df["existing_headcount"] = df["existing_permanent"] + df["existing_outsourced"]
    df["staffing_gap"] = df["required_headcount"] - df["existing_headcount"]
    return df

# optimize.py – permanent/outourced MIP (PuLP/CBC)
prob = pulp.LpProblem("EMX_permanent_outsourced_mix", pulp.LpMinimize)
perm_add = {s: pulp.LpVariable(f"perm_add_{s}", lowBound=0, cat="Integer") for s in stores}
outs_add = {s: pulp.LpVariable(f"outs_add_{s}", lowBound=0, cat="Integer") for s in stores}
mix_slack_hi = pulp.LpVariable("mix_slack_hi", lowBound=0)
mix_slack_lo = pulp.LpVariable("mix_slack_lo", lowBound=0)

for s in stores:
    prob += perm_add[s] + outs_add[s] >= gap[s]          # coverage

total_perm = pulp.lpSum([exist_perm[s] + perm_add[s] for s in stores])
prob += total_perm >= target_lo * approx_total - mix_slack_lo
prob += total_perm <= target_hi * approx_total + mix_slack_hi

prob += pulp.lpSum([perm_add[s]*C_perm + outs_add[s]*C_outs for s in stores]) \
    + 50000 * (mix_slack_hi + mix_slack_lo)
prob.solve(pulp.PULP_CBC_CMD(msg=False))
```

File inventory (all delivered, all executed):

File	Purpose	Lines
<code>config.py</code>	Every assumption, labelled by evidence status	~140
<code>data_gen.py</code>	Synthetic store/history/forecast generator (schema-matching, clearly labelled placeholder)	~110
<code>engine.py</code>	7-step capacity-conversion engine	~110
<code>baseline.py</code>	Reconstructed manual baseline	~65
<code>optimize.py</code>	Structural/peak classification + MIP allocator + hiring dates	~160
<code>backtest.py</code>	Time-based validation, no lookahead	~140
<code>scenarios.py</code>	10 scenario/sensitivity tests	~150
<code>explain.py</code>	Store-level explanation generator	~55
<code>run_all.py</code>	End-to-end orchestrator, produces every CSV/JSON/chart below	~130

Output artefacts produced by the executed run (all attached): stores.csv , history_actuals.csv , forecast_90d.csv , engine_daily_capacity_plan.csv , engine_hourly_capacity_plan_week1_sample.csv , store_hiring_recommendations.csv , optimisation_meta.json , example_explanations.json , backtest_detail.csv , backtest_summary.json , backtest_by_store.csv , backtest_by_archetype.csv , backtest_by_weekend.csv , scenario_results.csv , scenario_lead_time_delay.csv , scenario_mix_sensitivity.csv , scenario_outsourced_cost_sensitivity.csv , summary_charts.png .

13. Validation and Backtesting Results

13.1 Method (repeated for clarity — see also Section 3.2)

26 weeks (182 days) of synthetic history, evaluated store-day-by-store-day after a 28-day warm-up (1,848 store-days scored). At each evaluation date: (a) a **naïve seasonal forecast** uses only actuals strictly before that date (mean of the same weekday over the preceding ≤ 4 occurrences) — genuinely time-based, no lookahead; (b) the **reconstructed baseline** uses its own trailing-28-day average, also computed only from prior data; (c) the **“true” required headcount** is computed by running the same engine formula on the *actual* realised demand for that date — used only for scoring, never fed back into either forecast.

13.2 Headline results

Metric	Baseline	Engine	Target	Met?
Demand-capacity match accuracy (1 – mean abs % error, floored at 0)	80.4%	90.7%	95%	Not yet met — 4.3 points short on synthetic data
Mean absolute % staffing error	19.6%	9.3%	—	—
Total mismatch (over+understaffed) courier-hours	28,811	12,231	–20% vs. baseline	Exceeded — 57.5% reduction
Overstaffed courier-hours	13,187	5,711	—	–56.7%
Understaffed courier-hours	15,625	6,520	—	–58.3%
Scheduled labour cost/shipment (AED)	3.75	3.79	–0.50	Not met — engine is AED 0.05 <i>higher</i> on pure scheduled cost
Full-coverage cost/shipment, incl. emergency cost of uncovered shortfall (AED)	4.49	4.10	–0.50	Directionally met, not fully — AED 0.38 reduction

13.3 The cost result requires an honest explanation, not a headline number

The scheduled-labour-cost comparison (row 6) makes the engine look *slightly worse* than the baseline. This is not an error — it is the correct arithmetic consequence of the baseline being **chronically understaffed** (15,625 understaffed hours vs. the engine’s 6,520): a plan that recommends fewer courier-hours than are actually needed will always look cheaper on a naïve “cost of hours scheduled” basis, because it simply isn’t paying for hours it should be scheduling. The **full-coverage cost metric** (row 7) corrects for this by pricing the understaffed hours at an emergency-outsourced rate (representing what it actually costs, in a real operation, to plug an unplanned shortfall) — on that fairer basis the engine is cheaper by **AED 0.38 per shipment**, 76% of the way to the brief’s AED 0.50 target.

Exactly how the AED 0.50 target would be re-tested on real EMX data, per the brief’s own requirement to state shipment volume, baseline cost, proposed cost, time period, and included/excluded categories:

Element	Value in this synthetic test	What changes with real EMX data
Shipment volume	771,296 shipments over 26 weeks (synthetic)	Real EMX 26-week volume
Baseline labour cost	AED 2.89M scheduled / AED 3.46M full-coverage (synthetic)	Real payroll + real outsourced spend
Proposed labour cost	AED 2.93M scheduled / AED 3.16M full-coverage (synthetic)	Engine-recommended hours × real wage rates
Time period	26 weeks	Same, or first full pilot quarter
Included costs	Scheduled hourly labour + emergency-outsourced cost of shortfall	Add: overtime actually paid, hiring/onboarding cost, attrition cost — none are in this prototype’s cost model yet
Excluded	Fixed HR overhead, benefits beyond the “fully loaded” hourly rate,	Same exclusions recommended for the real test, to isolate the

costs	vehicle/fuel cost	labour-planning lever
Sensitivity range	Outsourced-cost-sensitivity table (Section 14) shows $\pm 35\%$ outsourced rate moves total optimiser cost by up to AED 511,056/year	Re-run with EMX's actual outsourced contract rates

13.4 Segment-level results (where the engine helps most and least)

Segment	Engine match accuracy	Baseline match accuracy	Improvement
Industrial/B2B stores	90.8%	66.4%	+24.4 pts
Urban-dense stores	91.3%	69.9%	+21.4 pts
Suburban stores	90.4%	88.2%	+2.2 pts
Mixed stores	90.6%	89.7%	+0.9 pts
Weekdays	90.8%	80.8%	+10.0 pts
Weekends	90.6%	79.3%	+11.3 pts

This pattern is itself a finding, not noise: the engine's advantage is largest exactly where the baseline's flat, single-productivity assumption is most wrong — industrial/B2B stores have the longest service+travel time per shipment (9.0 min/shipment vs. urban-dense's 4.6), so a flat 55-parcels/courier/day rule understaffs them badly (industrial_b2b baseline understaffed-hours: 9,522, essentially all of the segment's mismatch) while badly *overstaffing* dense urban stores (urban_dense baseline overstaffed-hours: 11,310). This is the concrete, measured version of the brief's own diagnosis — *“Do not rely on a single average productivity value across all stores unless the dataset proves that store differences are immaterial”* — and on this synthetic test, store differences are very much material.

Stores with the highest remaining engine error (worth flagging for early production attention, not concealing): AAN-01 (10.6%), AUH-03 (10.3%), SHJ-01 (10.1%) — all still outperform their own baseline error, so this is a “least improved,” not a “made worse,” list.

13.5 What the backtest does *not* prove

It proves the **relative** improvement of the engine's methodology over a flat-productivity baseline, on a schema-faithful synthetic dataset, under time-based (no-lookahead) validation. It does **not** prove the **absolute** headcount, cost or accuracy numbers describe EMX's real network — those depend entirely on the real store, roster, productivity and cost data substituted into `config.py` and `data_gen.py`'s replacement (a real ingestion pipeline).

14. Scenario and Sensitivity Analysis

All scenarios re-run the full engine (and, where noted, the optimiser) against the 90-day forward forecast — actual executed output, not estimated:

Scenario	Network avg required headcount	Network gap (heads)	Δ gap vs. base	Cost/shipment (AED)
Base case	116.9	43.9	—	3.82
Demand +10%	128.6	55.6	+11.7	3.82
Demand +20%	140.3	67.3	+23.4	3.82
Demand +30%	152.0	79.0	+35.1	3.82
Forecast deteriorates (plan to P95 band)	157.3	84.3	+40.4	3.82
Weekend demand spike (+25%)	126.1	53.1	+9.2	3.82
Absenteeism rises to 15%	128.5	55.5	+11.6	4.19
Productivity falls 15%	134.5	61.5	+17.6	4.39
New store added (cold start)	121.7	48.7	+4.8	3.83
Sustained campaign-level demand (Black Friday uplift, network-wide)	187.1	114.1	+70.2	3.82

Reading these correctly: cost-per-shipment stays flat (3.82) across pure demand-volume scenarios because required *hours* scale with shipments — the *headcount gap* grows, but cost per unit of work does not, which is the expected behaviour of a linear service-time model. Cost per shipment **does** rise when absenteeism or productivity worsens (4.19, 4.39) because the same shipment volume now needs more paid hours to deliver — this correctly isolates *efficiency* scenarios from *volume* scenarios, which the brief's KPI framework needs distinguished.

Permanent lead-time delay exposure (what happens if hiring takes longer than the 45–60 day assumption):

Extra delay	Effective lead time	Weeks of exposure	Courier-hours at risk
0 days	52.5 days	7.5 weeks	20,580
+15 days	67.5 days	9.6 weeks	26,460
+30 days	82.5 days	11.8 weeks	32,340

Mix-target sensitivity (already shown in Section 9.3): moving the target from 60% to 50% or 70% permanent changes annualised cost by ±AED 124,800 in this synthetic run — because permanent labour is assumed cheaper per hour, the 60/40 target is a flexibility choice, not a cost-minimising one (Section 9.3 explains why this matters).

Outsourced-cost sensitivity (re-costing the same fixed headcount plan under different outsourced market rates):

Outsourced rate multiplier	Rate (AED/hr)	Total annual cost	Δ vs. base
×1.00	27.00	3,061,760	—
×1.10	29.70	3,207,776	+146,016
×1.20	32.40	3,353,792	+292,032
×1.35	36.45	3,572,816	+511,056

15. KPI Measurement and Business-Case Framework

KPI	Exact formula	Owner (proposed)	Value in executed backtest
Demand-capacity match accuracy	$1 - \text{mean}(\text{recommended} - \text{true required} / \text{true required})$, floored at 0, by store/day	Planning lead	Engine 90.7% vs. baseline 80.4% (target 95%, not yet met)
Overstaffed store-hours	$\sum \max(\text{recommended} - \text{true required}, 0) \times \text{shift hours}$	Planning lead	5,711 hrs (engine) vs. 13,187 (baseline)
Understaffed store-hours	$\sum \max(\text{true required} - \text{recommended}, 0) \times \text{shift hours}$	Planning lead	6,520 hrs (engine) vs. 15,625 (baseline)
On-time-delivery risk	Not directly modelled in this prototype — requires real OTD-vs-staffing correlation data, which does not exist for EMX (Doc C p.38 gives the formula: $\text{delivered-within-commitment} \div \text{eligible delivered shipments}$)	Ops/CX lead	Unknown — flagged as the top data-integration priority for Phase 2 (Section 17)
Store closure risk	Binary flag: <code>gap_direction == "understaffed"</code> for N consecutive days beyond a severity threshold	Planning lead	0 store-closure-risk days flagged in the executed scenarios (no store's gap exceeded the closure-risk threshold used) — not a claim that EMX faces zero closure risk , only that this synthetic run did not trigger the flag
Courier utilisation	$\text{Productive paid time} \div \text{available paid time}$ (Doc A p.34)	Ops lead	Target 85% (Assumed); engine sizes headcount to hit this target by construction
Labour cost per shipment	$(\text{Scheduled hours} \times \text{hourly rate} + \text{emergency cost of uncovered shortfall}) \div \text{shipments}$	Finance lead	AED 4.10 (engine) vs. AED 4.49 (baseline), full-coverage basis — AED 0.38 reduction
Permanent/outsourced mix	$\text{Network permanent headcount} \div \text{network total headcount}$, post-recommendation	Workforce planning lead	64.8% realised vs. 60% target (within ±5pp band)
Emergency recruitment frequency	Count of hiring actions flagged “overdue” (need date already passed at time of decision)	Workforce planning lead	9 of 12 stores flagged overdue in the executed run — a direct measure of how far behind the <i>synthetic</i> existing workforce is from the <i>synthetic</i> forecast, illustrating the metric's mechanics, not an EMX finding
Hiring recommendation accuracy	To be measured post-pilot: recommended hires vs. hires that were still needed 90 days later	Workforce planning lead	Not measurable without a live pilot
		Workforce	

Planner override rate	Overrides logged ÷ total recommendations issued	planning lead	Requires the planner UI (Section 10.2, not yet built)
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Business-case discipline, per the brief’s explicit requirement: every AED figure in this report states its shipment volume (771,296, 26 weeks, synthetic), its baseline and proposed cost, its time period, its included/excluded cost categories, and its sensitivity range (Section 13.3, Section 14) — nothing is extrapolated to an annual EMX-wide saving, because doing so from synthetic data would misrepresent the finding as an EMX fact rather than a methodology demonstration.

16. Technical Architecture and Integration Requirements

Layer	Component	Notes
Source systems	Forecast system (existing EMX 3-month rolling forecast), HR/roster system, cost/payroll system	None of these were available for this exercise; engine is built against a defined schema (Section 11) so integration is a mapping exercise, not a redesign
Data pipeline	Scheduled ingestion (daily forecast refresh, weekly roster refresh) → validation (Section 3.2 checks) → engine run	Not built in this prototype; <code>data_gen.py</code> stands in for this layer and is designed to be swapped out cleanly
Data storage	Time-series store for forecast/actuals; relational store for store/workforce master data	Any standard warehouse (the prototype uses flat CSV/Parquet-ready DataFrames)
Model-training workflow	Weekly forecast refit; quarterly review of shrinkage/productivity assumptions against real HR/ops data	Section 7
Optimisation engine	PuLP + CBC (open-source MIP solver), proven to solve the 12-store network in well under a second — scales to hundreds of stores without a solver change	<code>optimize.py</code>
API/application layer	Engine outputs are flat, named-column CSV/JSON — directly consumable by a REST wrapper	Not built; recommended as a thin FastAPI layer around <code>run_all.py</code> ’s functions
Planner interface	Section 10.2 functional spec	Not built (out of scope per brief’s workforce-planning-only guardrail)
Authentication/access control	Standard SSO integration; role-based access (planner vs. approver vs. executive per Doc A p.2’s four decision classes)	Recommended, not built
Monitoring	Forecast error, backtest metrics, mix-band violations, solver status — all already computed by <code>backtest.py</code> / <code>optimize.py</code> and ready to feed a dashboard	—
Audit logs	Every recommendation should log its full input snapshot (forecast version, roster version, config version) alongside the output, so any recommendation can be reproduced exactly	Recommended design; the prototype’s <code>config.py</code> /data versioning approach (one file, one version) is a starting point
Model/data versioning	<code>config.py</code> and the data schema (Section 11) should be under version control; every engine run should tag its config version in its output	Recommended
Refresh frequency	Forecast: daily; capacity plan: daily; hiring/outsourcing recommendations: weekly review cadence	Matches Doc A p.19’s horizon table
Integration points	Forecast system, HR/roster system, payroll/cost system, recruitment/ATS system (for hiring-date automation)	—
Human approval	Per Doc A p.2’s decision-class framework: hiring and mix-band-breach recommendations should sit in the “system recommendation with planner approval” class, not “automated within approved constraints”	Design principle carried through Section 9’s rule layer

Ownership (proposed, since none is established in the PDFs): data quality → data engineering; forecast approval → demand planning; productivity standards → operations; workforce availability → HR/workforce planning; hiring decisions → workforce planning lead with finance sign-off above a cost threshold; outsourced capacity → operations/procurement; planner overrides → local store/regional ops manager; model monitoring → data science/analytics.

17. Implementation and Pilot Plan

Phase	Inputs required	Deliverables	Validation gate	Main dependency	Principal risk
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1. Data validation & baseline creation	Real EMX shipment history (≥6 months), real roster, real cost rates	<code>config.py / data_gen.py</code> replaced with real ingestion; reconstructed baseline re-run on real data	Baseline reproduces a documented current planning number within an agreed tolerance, or the gap is explained	Access to EMX data owners	Real data may reveal the Assumed shrinkage/productivity parameters are badly wrong — expected and desired at this stage
2. Store-level capacity model	Phase 1 outputs + real store archetypes/productivity	Engine formulas re-calibrated per store or per real archetype cluster	Engine's store-level error beats the baseline's on a real held-out period (mirroring Section 13's method)	Sufficient per-store history (engine currently assumes enough data for per-store or per-archetype productivity; true cold-start stores use the archetype prior, as tested in Section 14)	Productivity data may not exist at store grain yet — same gap the PDFs flag for EMX overall
3. Permanent/outsourced optimisation	Real cost rates, real lead times (confirm the brief's 45–60/5–10 days against EMX's actual recruitment/outsourcing contracts)	MIP re-solved on real structural gaps	Solver stays "Optimal" (not infeasible) across at least 4 consecutive weekly runs	Real 60/40 policy confirmation (Section 9.3's definition needs sign-off)	If real lead times are longer (Section 14's delay scenario), current structural-gap-confirmation logic (4 weeks) may need lengthening
4. Hiring recommendation engine	Phase 3 + recruitment/ATS integration	Automated hiring-date generation feeding the recruitment pipeline	Hiring dates cross-checked against at least one real recruitment cycle	ATS/recruitment system access	Recommendation accuracy (recommended hires vs. hires still needed 90 days later) can only be measured after a full lead-time cycle has elapsed
5. Planner prototype	Phases 1–4	Thin UI over Section 10.2's spec (override + audit trail is the minimum viable feature)	Planner can action a real recommendation end-to-end in a usability test	UI/product resourcing (not costed in this exercise)	Adoption risk if explanations aren't trusted — mitigated by Section 10.1's real-model-output explanation design
6. Controlled pilot	2–3 stores, one full permanent hiring cycle (≥60 days)	Live comparison: engine recommendation vs. actual planner decision vs. actual outcome	Engine's forecast-to-capacity accuracy beats the store's own historical planning accuracy	Willing pilot stores, planner buy-in	Small-sample noise — 2–3 stores over one cycle is not statistically strong; treat as qualitative validation
7. Performance validation	Pilot data	Full KPI table (Section 15) computed on real pilot data	Match accuracy, mismatch-hour reduction, and cost-per-shipment targets (Section 15) assessed against real numbers, not synthetic ones	Pilot completion	Results may differ substantially from Section 13's synthetic backtest — expected, and the reason a pilot precedes wider rollout
8. Wider deployment	Phase 7 sign-off	Network-wide rollout, monitoring dashboard live	KPI targets sustained for 2 consecutive quarters	Executive approval, ongoing data pipeline ownership	Scale-up of the MIP is not a risk (CBC solves this network in <1 second; hundreds of stores remain tractable)

18. Risks, Limitations, and Controls

Risk/limitation	Severity	Control

Every operational parameter is currently a placeholder (Section 4)	High — this is the central limitation of the entire exercise	<code>config.py</code> isolates every Assumed value in one auditable file; Phase 1 of Section 17 exists specifically to replace them before any number in this report is used for a real decision
Reconstructed baseline may not match EMX's actual current practice, which remains genuinely Unknown	Medium	Explicitly labelled "reconstructed" throughout; Phase 1 should attempt to document the real current method alongside the data validation work
Hourly demand curve and peak/structural split are the weakest-evidenced parts of the engine (Sections 8.1, 9.1)	Medium	Flagged explicitly as top candidates for early real-data substitution; peak/structural split should move to a volatility- or uncertainty-band-based method once real daily history exists
60/40 mix definition is an interpretation, not a confirmed EMX policy detail	Medium	Section 9.3 states the interpretation and the alternative readings explicitly; needs policy sign-off before production use
MIP mix-constraint linearisation is an approximation (Section 8)	Low-medium	Realised ratio is always recomputed and reported post-solve, so any drift from the target band is visible, not hidden
Synthetic backtest cannot prove real-world forecast accuracy	High, by design	Section 13.5 states this explicitly; a real backtest (Phase 1) is a hard prerequisite for any production cost claim
No OTD/service-level data exists to validate the "95% on-time delivery" and "zero store closures" targets against staffing decisions	High	Flagged as Unknown in Section 15, not fabricated; Phase 2/6 should integrate real OTD data as soon as it's available
Emergency-outsourced-cost assumption (×1.35 premium) driving the full-coverage cost result (Section 13.3) is itself Assumed	Medium	Section 14's outsourced-cost-sensitivity table shows how much the headline AED 0.38 figure moves under alternative rate assumptions
No planner UI/override/audit trail is built yet	Medium	Explicitly scoped out per the brief's workforce-planning-focus guardrail; Section 10.2 provides the functional spec so this is a scoped Phase 5 build, not a surprise gap
Overtime is modelled as a cost parameter but not yet exercised by any scenario	Low	Present in <code>config.py</code> ; recommended as an explicit scenario in Phase 2/3 once real shift/overtime rules are known

19. Final Assessment Against the Judging Criteria

Requirement	Status
Converts shipment forecasts into defensible staffing and hiring decisions	Met — Sections 8–9, executed end-to-end
Reflects store-level operational differences	Met — archetype-differentiated service/travel time; demonstrated to matter most exactly where expected (Section 13.4)
Incorporates workforce availability and lead times	Met — Section 9, hiring-date logic uses the brief's exact lead times
Produces explainable recommendations	Met — Section 10.1, generated from live model state, brief's format followed exactly
Demonstrates measurable comparison with a manual baseline	Met — Section 13, reconstructed and explicitly labelled as such
Uses AI and optimisation meaningfully	Met — hierarchical forecasting design (Section 7) + MIP allocation (Section 8), not "ordinary formulas dressed as AI"
Is technically reproducible	Met — all code executed in this session; files delivered; results are actual output, not illustrative
Can realistically be piloted within EMX	Met, with a hard precondition — Section 17's Phase 1 (real data substitution) must complete first; the architecture itself requires no redesign to accept real data
Makes assumptions and missing data visible	Met — Section 4 and <code>config.py</code> ; this is arguably the most load-bearing section of the whole report, given Section 3's finding that no EMX operational figure exists in any source document
Avoids fabricated performance claims	Met — Section 13.3 explicitly reports where a target was <i>not</i> met (scheduled cost, 95% match accuracy) alongside where it was exceeded (mismatch-hour reduction); Section 15 states OTD/closure-risk KPIs as Unknown rather than invented
Directly addresses the stated KPIs and judging criteria	Met — Section 15

Where this submission is intentionally incomplete, stated plainly: the planner UI is a specification, not a build (Section 10.2); OTD/closure-risk validation requires data that does not exist yet (Section 15); and every absolute number is a demonstration of methodology on synthetic data, not an EMX performance claim (Sections 4, 13.5). These are the honest boundaries of what can be delivered without real EMX data, stated up front rather than discovered by a reviewer.

20. Evidence Register

#	Claim	Document	Page	Status
1	No EMX-specific revenue, volume, headcount, fleet, or capacity figures are publicly available	Doc B	3	Verified
2	No public employee/courier count, org chart, or staffing level was identified	Doc B	36	Verified
3	Public evidence does not establish EMX workforce scale, cost-to-serve, or service-level performance	Doc C	1	Verified
4	Blueprint assumes a generic large multi-hub carrier “because no company-specific operating data has been supplied”	Doc A	1	Verified
5	“Limited operating-scale disclosure” confirmed as a disclosure gap, with market-position implications	Doc B	41	Verified
6	Forecast-horizon-by-decision table (same-day: hrs–2 days; shift/subcontractor booking: 1–6 weeks; vehicle/facility: 1–6 months; line-haul contracts: 3–18 months; hub investment: 2–10 years)	Doc A	19	Verified
7	“Capacity forecasting” engine: demand forecast + productivity + absence distributions → required resources by interval, output as “scenario bands, not single-point values”	Doc A	17	Verified
8	“Demand forecasting” engine: hierarchical time-series, gradient boosting, event regressors	Doc A	17	Verified
9	Driver utilisation = productive paid time ÷ available paid time	Doc A	34	Verified
10	“Increase capacity utilisation” objective, measured as parcels per vehicle/hub/labour hour	Doc A	9	Verified
11	“Forecast-led labour, vehicle and facility-capacity planning” listed as a highest-value capability	Doc A	3	Verified
12	On-time delivery = delivered within product commitment ÷ eligible delivered shipments	Doc C	38	Verified
13	Parcels per labour hour = completed eligible parcel tasks ÷ paid direct labour hours	Doc C	38	Verified
14	Stops per hour = completed stops ÷ route operating hours	Doc C	38	Verified
15	Design principle: deterministic rules for safety/legal; optimisation for allocation/routing/scheduling; ML for uncertain outcomes (demand, travel time, etc.)	Doc A	16	Verified
16	“Workforce planning” listed as a “Should have” (buy/configure) MoSCoW capability, not a “Must have” build item	Doc A / Doc C	39 / 43	Verified
17	Core data domain “Workforce”: driver, courier, skill, shift, break, training, performance	Doc A	23	Verified
18	Workforce-management integration: shift, absence, skill and hours data, bidirectional	Doc A	28	Verified
19	Material seasonal peaks named qualitatively around Ramadan, Eid, Black Friday, year-end (not quantified)	Doc A	1	Verified (qualitative only)
20	Four decision classes with recommended decision modes (automated / planner-approved / rules-based+human / executive scenario modelling)	Doc A	2	Verified
21	60/40 permanent/outsourced mix target	Brief	—	Verified-Brief (no PDF corroboration found on exhaustive keyword search)
22	45–60 day permanent / 5–10 day outsourced lead times	Brief	—	Verified-Brief
23	Target KPIs (95% match accuracy, 20% mismatch reduction, 95% OTD, AED 0.50/shipment cost reduction)	Brief	—	Verified-Brief
24	All shift lengths, absence rates, productivity values, cost rates, and hourly demand shapes used in the prototype	config.py	—	Assumed — no PDF or dataset source exists

25	All backtest/scenario results reported in Sections 13–15	outputs/*.csv, *.json	—	Derived — actual output of executed code on synthetic, labelled data
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Full prototype code, all output CSVs/JSON, and the summary chart are provided as separate downloadable files alongside this report.